Intensity clipping for 3D lesion segmentation

Does the observed improvement justify a larger replicated study?

Research update

Public-safe synthetic evaluation summary



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A controlled comparison on 18 validation volumes

Same baseline and candidate setup

- Split, model architecture, training schedule and post-processing.

Only intended difference

- Test-time intensity clipping at the 1st and 99th percentiles.

Public-safe synthetic evaluation summary prepared for release acceptance.

This is a small validation study, not a final benchmark or clinical claim.

Observed metrics favor intensity clipping

Metric	Baseline	Intensity clipping
Mean Dice	0.742	0.758
Small-lesion Dice	0.421	0.469
HD95	18.4 mm	17.9 mm

The result supports replication, not a superiority claim.

No confidence interval or repeated-seed estimate is available.

The motion-corrupted case remains unresolved

Observed failure

One motion-corrupted volume still has disconnected false positives under both methods.

Interpretation limits

- No confidence interval or repeated-seed estimate is available.
- The observed improvement may depend on this small split.

The result supports replication, not a superiority claim.

Approve a three-seed replication?

Proposed next experiment

1. Repeat the comparison across three fixed seeds.
2. Pre-register the same preprocessing and split policy.
3. Report uncertainty for mean Dice, small-lesion Dice and HD95.
4. Audit the motion-corrupted case separately instead of excluding it.

Decision request

Approve the three-seed replication before changing the model architecture.